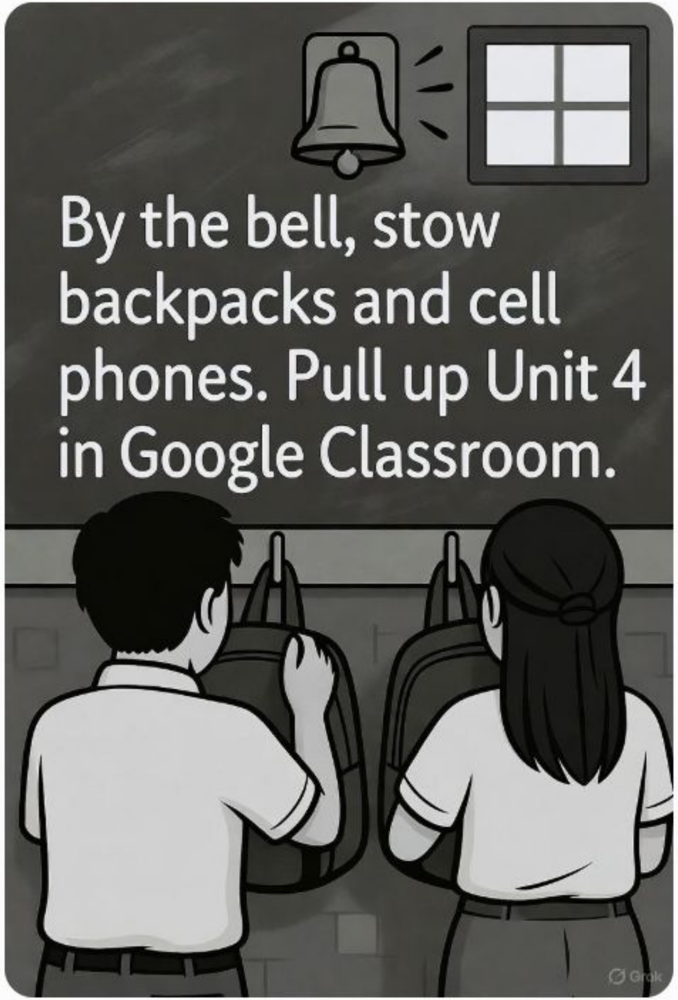
Unit 4: Foundations in Safety & Robotics



Learning Objectives

1. Review Vicksburg Robotics Lab Rules
2. Describe key OSHA regulations that apply in a machine/robotics lab.
3. Identify and select appropriate Personal Protective Equipment (PPE).
4. Read and interpret a Safety Data Sheet (SDS) for common shop chemicals.

Lab Rules

Safety First:

- Always wear appropriate personal protective equipment (PPE) such as safety glasses, closed-toe shoes, and, when necessary, gloves or other gear as directed by the instructor or mentor.
- Secure long hair and loose clothing to prevent entanglement in machinery or tools.
- Follow all safety protocols for tools, machines, and equipment (e.g., milling machines, drill presses, CNC machines) as outlined in training sessions.
- Report any hazards, spills, or equipment malfunctions to a mentor or instructor immediately.
- No horseplay, running, or disruptive behavior in the lab/shop.

Lab Rules

Tool and Equipment Use:

- Only use tools and equipment (e.g., hand tools, power tools, 3D printers) after receiving proper training and authorization.
- Inspect tools and machines before use; do not use damaged or faulty equipment (i.e., missing guard, damaged blades, exposed wire, damaged or missing shields, or Locked/Tagged Out).
- Follow proper procedures for setup, operation, and shutdown of equipment, including calibration and maintenance checks.
- Return tools to their designated storage areas after use and clean workspaces.

Lab Rules

Respect the Workspace:

- Keep work areas clean, organized, and clutter-free to ensure safety and efficiency for all team members.
- Recharge tool batteries after use to keep equipment ready for the next project.
- Do not touch or disturb others' projects, respecting the hard work and creativity of your teammates.
- Store materials (e.g., metals, plastics, electronics) properly to prevent damage or clutter. Store material flat on floor or in racks to prevent unsafe conditions.
- Do not leave projects or tools unattended without securing them.
- Label personal or team projects clearly to maintain organization and avoid confusion.
- No food or drink is allowed in the lab or shop except in designated areas to maintain cleanliness and protect equipment.

Lab Rules

Lab Access and Security:

- Only access the lab during designated hours or with explicit permission from a mentor or instructor.
- Do not allow unauthorized individuals into the lab.
- Ensure the lab is locked and equipment is secured when leaving.
- Report any observed unauthorized or improper use of the lab to the instructor immediately.

Lab Rules

Emergency Procedures:

- Know the location of fire extinguishers, first aid kits, and emergency exits.
- Follow evacuation procedures during drills or emergencies.
- Report injuries, no matter how minor, to a mentor or instructor immediately.

Learning Check

What is the primary purpose of the Vicksburg Robotics Lab/Shop Rules?

a) To encourage competition

$\boxed{\text{b) To ensure a safe and productive environment}}$

c) To limit tool access

d) To assign homework

Learning Check

What should you do if you see a damaged tool guard?

a) Use the tool anyway

b) Report it to the instructor immediately

c) Ignore it if it looks minor

d) Fix it yourself

Learning Check

Which of these is a sign of an unsafe shop condition?

a) Clean workspace

b) Unattended running machine

c) Properly stored tools

d) Worn PPE

Learning Check

What is the main purpose of Lockout/Tagout (LOTO) in the shop?

a) To speed up tool maintenance

b) To prevent accidental startup of equipment during maintenance

c) To test equipment power

d) To lock tools for storage

Think back—what’s the most dangerous situation you’ve seen in a shop or lab? How could it have been prevented?

OSHA Overview

Occupational Safety and Health Administration

OSHA Overview

You Have Rights at Work

You have the right to:

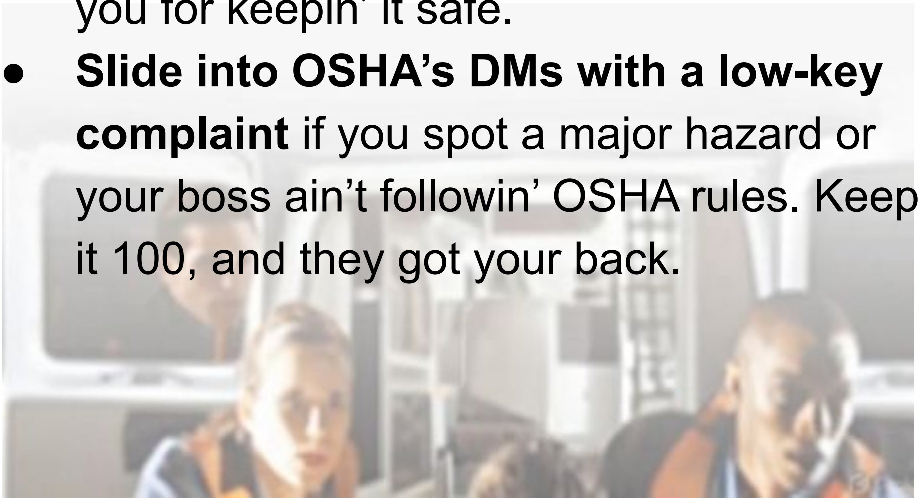
- Work in a safe place.
- Receive safety and health training in a language that you understand.
- Ask questions if you don't understand instructions or if something seems unsafe.
- Use and be trained on required safety gear, such as hard hats, goggles and ear plugs.
- Exercise your workplace safety rights without retaliation or discrimination.
- **File a confidential complaint** with OSHA if you believe there is a serious hazard or that your employer is not following OSHA standards.

OSHA Overview (Shameless Gen Alpha Translation)

Yo, You Got Rights in the Shop, Fr!

You got the drip to:

- **Work in a safe spot.** No sketchy vibes—your workplace gotta be secure, no cap.
- **Get safety and health training that slaps in your language.** They gotta break it down so you vibe with it.
- **Ask Qs if you're lost or somethin' feels off.** Don't sleep on unclear instructions—speak up, bet.
- **Use and learn how to rock safety gear like hard hats, goggles, and ear plugs.** Stay protected, fam—it's a must.
- **Flex your safety rights without catchin' shade or drama.** No one can come for you for keepin' it safe.
- **Slide into OSHA's DMs with a low-key complaint** if you spot a major hazard or your boss ain't followin' OSHA rules. Keep it 100, and they got your back.



OSHA Overview

Your Employer Has Responsibilities

Your employer must:

- Provide a workplace free from serious recognized hazards and follow all OSHA safety and health standards.
- Provide training about workplace hazards and required safety gear.*
- Provide training about workplace hazards and required safety gear.*
- Tell you where to get answers to your safety or health questions.
- Tell you what to do if you get hurt on the job.

**Employers must pay for most types of safety gear.*

OSHA Overview

OSHA Overview

What Is OSHA?

- **Mission:** “Ensure safe and healthful working conditions by setting and enforcing standards.”
- **Scope:** Covers all private-sector employers, including educational labs and shops.
- **OSHA’s 3 Key Functions:**
    1. **Standard-setting** (e.g. machine guarding, PPE)
    2. **Inspections & Enforcement**
    3. **Training & Outreach**

OSHA Overview

Key Standards for Our Lab

Standard	Citation	Brief Description
Machine Guarding	29 CFR 1910.212	Requires guards on all points of operation (belts, gears)
Walking-Working Surfaces	29 CFR 1910.22	Keep floors clean, dry, clear of tripping hazards
Hazard Communication	29 CFR 1910.1200	Labels + Safety Data Sheets for all chemicals

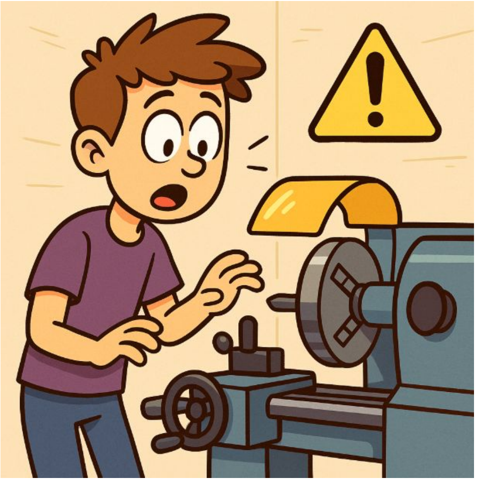
OSHA Overview

Why It Matters to Us

- **Prevents Injuries:** Cuts, amputations, slips, chemical exposures

- **Keeps Us Legal:** Avoid fines (up to $13,653 per violation)

- **Builds Credibility:** Demonstrates professionalism to administrators and sponsors



OSHA Overview - Recordable Injuries

Injuries that require medical attention beyond First Aid

Work-related injuries or illnesses that require medical attention *beyond* first aid are recordable. Consider the example of an employee getting debris in their eye. The physician on site uses eye flush solution and does not administer pain medication. The employee is sent home for the rest of the day. This *is not* a recordable incident because irrigating the eye is considered first aid.

However, if the debris caused a scratch on the worker's eye and they receive a medicated drop to administer on a daily basis, that goes beyond first aid and, therefore, *is* a recordable incident.

This can be nuanced, which is why OSHA clearly defines first aid in 29 CFR 1904.7(b)(5)(ii).

OSHA Overview - Reportable Injuries

Fatalities, in-patient hospitalization, amputation or loss of an eye

A recordable incident becomes a reportable incident when it meets two specific criteria: 1) it causes a fatality, or 2) it causes injuries that require in-patient hospitalization or that result in amputation or loss of an eye.

For reportable incidents, it’s important to know these OSHA mandates:

A fatality must be reported to OSHA within eight hours.
An in-patient hospitalization, amputation or loss of an eye must be reported within 24 hours.

Additionally, a fatality that occurs within 30 days of the incident, or an in-patient hospitalization, amputation or loss of eye that occurs within 24 hours of the work-related incident must be reported.

Learning Check

What should you do if you don’t understand a safety instruction?

a) Proceed anyway

b) Ask the instructor for clarification

c) Guess the meaning

d) Ignore it

Learning Check

What is an example of personal accountability in the shop?

a) Leaving tools out

b) Reporting a hazard you see, even if you didn’t create it

c) Using a tool without training

d) Ignoring safety rules

Learning Check

How should you report a safety concern?

a) Tell all of your friends and at least three strangers

b) Notify the instructor immediately

c) Write it in your journal

d) Do nothing

Quick Check – True or False?

T/F: You must have an SDS on file for every chemical in the shop.

Quick Check – True or False?

T/F: OSHA only applies to big industrial plants, not school labs.

Quick Check – True or False?

T/F: A machine without a guard is an immediate OSHA violation.

Lockout/Tagout (LOTO) Fundamentals

Definition

- **Lockout/Tagout:** The process of isolating and securing all sources of hazardous energy (electrical, pneumatic, hydraulic, mechanical) before maintenance or servicing.

Lockout/Tagout (LOTO) Fundamentals

Scope

- **Students:** Will not lock, unlock, or service machines.
- **Students**: Will recognize locked-out machines and not use them.

Quick Check – True or False?

T/F: You may remove a lock and tag to return a machine to service as long as no one is still working on it.

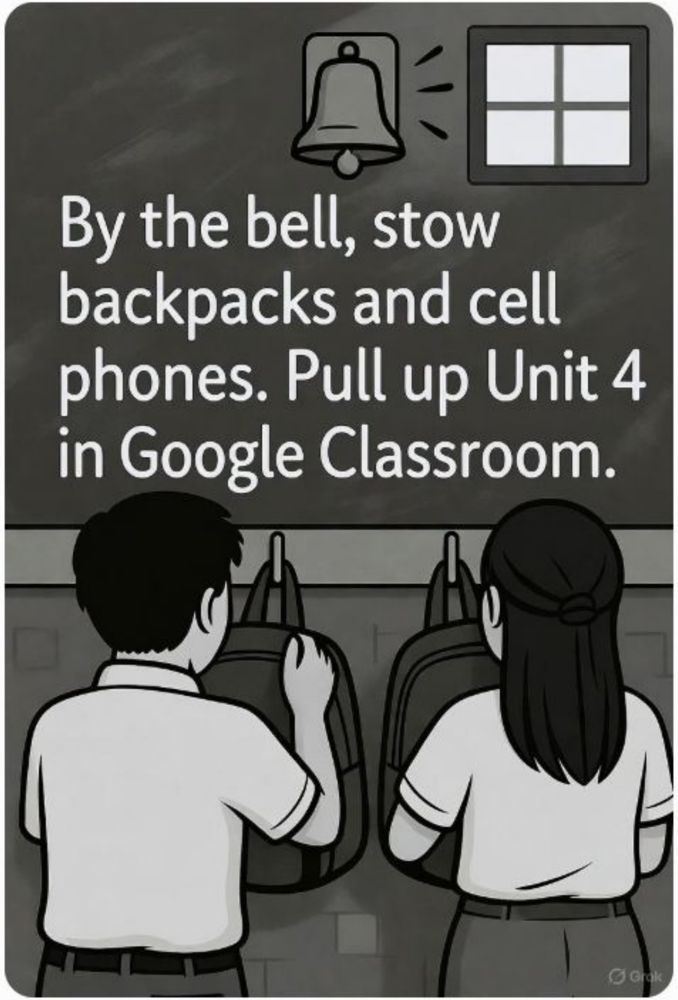
Only the person who placed the lock may remove it, thus the tag

What are the hazards in our shop?

- General shop environment
- Tool specific
- Chemical

The Worst Week of Roller Coaster Accidents

Today's Learning Objective:
Safety by Design, Identify key safety features from the Lac Megantic and Matthew Henne incidents and propose one engineering control or safety design feature to prevent similar risks.



Engineering Controls: Safety by Design

Physical solutions to reduce risks (e.g., barriers, sensors)

Prevent accidents before human error (like Matthew Henne’s)

Safe tool/equipment operation

How Controls Could’ve Saved Matthew

Interlocked doors: Stop ride if opened

Motion sensors: Halt if someone’s near

Barriers: Block access to moving parts

Applying Controls Here

Plasma CNC: Water table to arrest sparks. Ventilation for fumes

Table Saw: Blade guards (pre-installed) and Saw Stop Technology

Lac-Mégantic: A Night of Tragedy

July 6, 2013: Runaway train derails, 47 deaths

72 oil tank cars explode in downtown

Impact: 30+ buildings destroyed, river contaminated

What Went Wrong at Lac-Mégantic

Root Causes Unveiled

- Too few handbrakes (5 vs. 17 needed)
- Fire disabled air brakes
- Single-person crew, poor maintenance

Tom Harding applied 5 brakes—why wasn’t that enough?

Safety Culture Matters

Multi-person crews now required

Better brake checks and backups

Tie to our shop: Never leave tools unattended